

Introduction

Project Topic

Natural Language Processing has made significant progress in sentiment analysis of text, however, speech emotion recognition (SER) still has a long way to go. This is because in addition to linguistic qualities, like words, syntax, and cultural or language specific meaning, there are also paralinguistic qualities, like vocal characteristics and prosody, that can influence the interpretation of speech. Text and speech can also sometimes complement each other to enhance meaning or can contradict each other, where the emotional affect differs from the context derived from the spoken words. The quality of the speech can be degraded by external factors like noise, audio-recording quality, sampling rate, and environmental acoustics. Furthermore, while a variety of models and data processing techniques have differing strengths in SER analysis, speaker specific bias is high, which has potential to reduce model validity.

Background

Speech is considered a "variable-length transmission that conveys both information and emotion, making it necessary to extract both global and local features using appropriate technique" (Ezzameli & Mahersia, 2023). The acoustic, non-verbal properties of vocal communication are known as paralinguistic features. These non-linguistic speech features are broadly categorized based on their temporal structure as being either segmental (short-time) or suprasegmental (long-time, entire utterances) based on low-level descriptors (LLDs) and statistical measures derived from LLDs known as functionals (Anagnostopoulos et al., 2012). LLDs represent prosodic and spectral features of speech, with earlier research focusing on suprasegmental prosodic aspects such as fundamental frequency (i.e. pitch) and intensity and more recent research incorporating elements

like voice quality, such as suprasegmental features like jitter and shimmer, and spectral features, which include segmental features, such as Mel frequency cepstral coefficients (MFCCs) (Anagnostopoulos et al., 2012; Schuller et al., 2009). Common functionals applied to LLDs include extreme values, means, and moments (e.g. standard deviation), as well as many others (Anagnostopoulos et al., 2012; Schuller et al., 2009).

Once data is preprocessed and feature selection completed, there are a variety of classifiers used for SER tasks to model emotional states. While choice of classifier does impact performance, there is no uniform standard for evaluation and no best method established. Common models include Gaussian Mixture Models (GMMs), Hidden Markov Models (HMMs), Support Vector Machines (SVMs), Artificial Neural Networks (ANNs), Decision Trees, and k-Nearest Neighbor distance classifiers (kNNs) used as single classifiers, as well as hybrid multi-classifier systems (Anagnostopoulos et al., 2012). Focusing on specific models that were explored in class, SVMs have strengths in generalizability, ANNs like Multi-Layer Perceptron (MLP) and deep neural networks like Convolutional Neural Network (CNN) are efficient in modeling nonlinear mappings and handling smaller training data, and Decision Trees and ensemble classifiers like the Random Forest (RF) classifier handle large numbers of input features without relying on data sparseness and feature detection (Anagnostopoulos et al., 2012).

For example, El Hizabri et al., (2026) has shown success using an MLP classifier with a 41 neuron input layer matching the feature vector, two dense layers of 256 and 128 neurons, a Rectified Linear Unit (ReLU) activation function, and Adam optimizer with the feature vector smoothed by a Kalman Filter to reduce frame-level jitter, the model was able to achieve 87% accuracy on the RAVDESS dataset,

suggesting that Kalman smoothing enhanced generalizability (El Hizabri et al., 2026). However, each of these classifiers have weaknesses that impact performance in specific scenarios. With SVMs, there is no optimal strategy for choosing the kernel function and achieving perfect feature separation can lead to overfitting (Anagnostopoulos et al., 2012). MLPs may have better performance for nonlinearly separable data, but hyperparameters, like design of the neuron arrangement and interconnectedness, activation function selection, number of training epochs, learning rate, and validation methods, all need to be chosen carefully as they are often optimized for specific models, thus lacking external validity (Anagnostopoulos et al., 2012). Lastly, Random Forest classifiers can quickly become uninterpretable due to risk of overfitting from poor tree pruning (Anagnostopoulos et al., 2012).

Furthermore, there are still significant challenges in handling SER tasks across multiple different speakers and spontaneous novel speech (Schuller et al., 2009). To address this, as well as the issue of data sparseness, Schuller et al. (2010) proposed exploring evaluation of multi-lingual cross-corpora datasets to enhance independence between training and testing sets. By applying speaker and corpus normalization to handle variability across individual speaker communication quality and corpus environmental recording quality, the research found similarities across corpora, particularly following speaker normalization, leading to the best results using supra-segmental feature analysis for an SVM classifier (Anagnostopoulos et al., 2012; Schuller et al., 2010). The research noted that limitations to cross-corpora datasets include variability in cultural aspects of emotional expression and that the speech used for this study had limited linguistic content (Schuller et al., 2010).

One novel effort to improve data robustness despite the limitations in SER dataset size and generalization has been the wav2vec 2.0 model. Wav2vec 2.0 uses a multi-layer CNN to convert raw speech audio into latent speech representations and then feeds this into a transformer to capture physical attributes and emotional variations in speech without the need for manual feature design and transcribed audio data (Baevski et al., 2020). Research by Wang and Yang (2025) demonstrated successful application of the wav2vec 2.0 model where the authors saw that using the wav2vec 2.0 model for feature extraction followed by a Neural Controlled Differential Equation classifier and MLP achieved accuracies of 73-74% with convergence reached after one epoch of training.

Experimental Question

Due to challenges in speech emotion recognition, this project focuses on interpreting raw acoustic data from the Toronto Emotional Speech Set (TESS) to address how different model approaches for classifying emotion in speech compare in their pros and cons. Through identification of the strengths and weaknesses of different approaches, this project identifies the generalizability of different models to determine the extent to which feature extraction techniques, like temporal smoothing via Kalman Filter or transformer-based acoustic embeddings via wav2vec 2.0, reduces speaker bias in cross-speaker emotion classification compared to traditional static spectral features, like MFCC.

Methods

Description of Data

The Toronto Emotional Speech Set (TESS) dataset consists of 2,800 NU-6 Test audio recording files consisting of 200 target words accompanied by the carrier phrase "Say the word ___" in seven distinct emotional states (happy, disgust, neutral, fear, angry, pleasant surprise, sad) spoken by a two actresses, young

and old (yaf, oaf) recorded at 24.414 kHz sampling rate (Pichora-Fuller & Dupuis, 2020). TESS was obtained from Kaggle. It was chosen in order to isolate paralinguistic aspects of SER. File names were standardized to lowercase and to account for naming discrepancies, and then mapped to speaker and emotion as a Pandas DataFrame.

To test generalization and avoid data leakage, data was split along the speaker so that training data was from the older adult female (oaf) and testing data was from the younger adult female (yaf). Then extracted features were normalized using StandardScaler fit on the training data and then applied to both train and test data prior to classification for all approaches.

Data Preprocessing and Analysis Process

Approach 1: MFCCs (Mean & Standard Deviation) + Support Vector Machine

Original sampling rate was maintained due to the high quality, clean audio files. Librosa was used to extract 20 MFCCs per audio file to capture timbre and spectral envelope. This was then used to calculate static spectral features via mean and standard deviation, which were stored in a 40-dimension feature vector using NumPy. The training dataset was then provided to the C-Support Vector Classifier (SVC) with hyperparameters of RBF kernel and optimal C value of 1 as identified by hyperparameter tuning.

Approach 2: Kalman Smoothing of MFCCs & Deltas + Random Forest

Similar to the previous approach, the sampling rate was maintained and 20 MFCCs per audio file were extracted. MFCCs were smoothed individually using a Linear Kalman Filter from FilterPy to reduce noise in order to emphasize the broader emotional spectral contour. To smooth frame-level jitter, the Kalman Filter was set with process noise (Q) at

0.1 and optimal measurement noise (R) at 100 as identified by hyperparameter tuning to obtain a balance between temporal continuity and noise suppression. Then the respective Deltas (Δ), which is the first derivative of the MFCC and represents the temporal component velocity as the rate the coefficients change between frames, were calculated from the MFCCs. This yielded an 80-dimension feature vector consisting of the mean and standard deviations for the MFCCs and Deltas. The training dataset was then processed by a Random Forest Classifier with 100 trees in the forest, max tree depth of 5, minimum 20 samples at a leaf node, and random state of 42, which was identified as optimal based on various sets of hyperparameters explored.

Approach 3: Kalman Smoothing of MFCCs & Deltas + Multi-Layer Perceptron

The same extracted data from the second approach was used with the only difference being that an MLP classifier was trained. In this model, three hidden layer sizes became gradually smaller and more compressive (128, 64, and 32 neurons), the ReLU activation function, Adam optimizer, alpha of 0.01 to prevent overfitting, default initial learning rate of 0.001, 500 max iterations, and random state of 42 were used, with early stopping enabled and 10% of the training data reserved for validation monitoring.

Approach 4: wav2vec 2.0 + Multi-Layer Perceptron

The same MLP classifier setup from the third approach was repeated, however feature extraction was changed to a deep learning framework to replace manual feature extraction. Audio files were resampled at 16kHz using Librosa and then passed through the basic wav2vec 2.0 architecture from the Transformers library to extract contextual embeddings; then

averaged using Torch into a single dimension 768-element feature vector.

Approach 5: wav2vec 2.0 with Layer Extraction & Pyramid Pooling + Multi-Layer Perceptron

Further exploration of basic wav2vec 2.0 involved disabling drop-outs to reduce the risk of overfitting and enabling return of hidden states to measure MLP with extraction of layer 8 alone and layers 6 and 8 combined. The middle layers of wav2vec 2.0 provide the most prosodic and phonetic information with layer 6 focusing more on acoustic features like spectral envelope and layer 8 focusing more on prosodic features like pitch trajectory. This was chosen to eliminate noise from layers that could be biasing the model, since the data has redundant linguistic context and generalizing too strongly on acoustics could unintentionally cause the model to learn one speaker's voice.

Pyramid pooling was then performed, where the embedding of the layer(s) were divided row-wise along the temporal axis into 1, 2, 4, and 8 equal segments with each chunk then averaged and standard deviation obtained, and then all arrays were concatenated as rows. This resulted in a 23,040-dimension feature vector for layer 8 alone and a 46,080-dimension feature vector for layers 6 and 8 combined, where $\text{Total Dimension} = \# \text{ Layers Extracted} * \text{Sum of the Set of Bucket Sizes} * \# \text{ of Statistical Measures Used} * \text{Dimension of the wav2vec 2.0 Hidden State Layers}$. For example, for the ensemble approach: $46,080 = 2 * (1 + 2 + 4 + 8) * 2 * 768$.

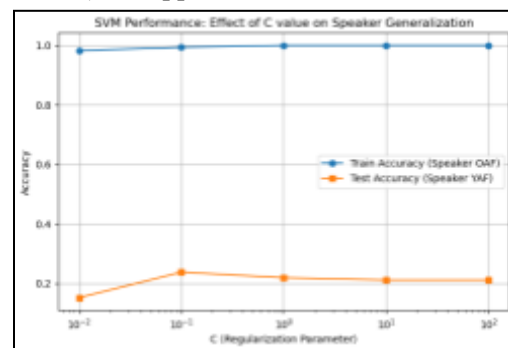
Note that an additional attempt at data processing not documented in the code or explored further due to worsened performance was Principal Component Analysis (PCA) on the scaled features. PCA acts as a linear filter to reduce spectral jitter. Performance that was only slightly better than random chance with introduction of PCA likely reflects the importance of preserving the nonlinear patterns

from the wav2vec transformer in emotion representation.

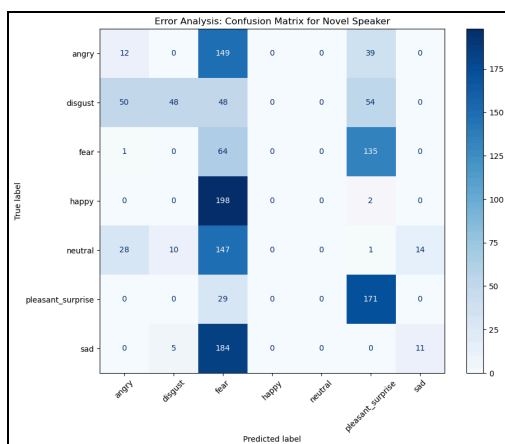
The MLP classifier was similar to before, with the only changes being reducing max iterations to the default and increasing alpha to 0.05 to apply stronger regularization to fight speaker overfitting due to more features. Hidden layer sizes were also similarly increased to 512, 256, and 128 neurons for the model using only layer 8 and 1024, 512, and 256 neurons for the model using layers 6 and 8. Additionally, during model development, various alphas, learning rates, and thresholds for number of iterations without change were also measured individually with the best performance kept.

Results

Approach 1: MFCCs (Mean & Standard Deviation) + Support Vector Machine



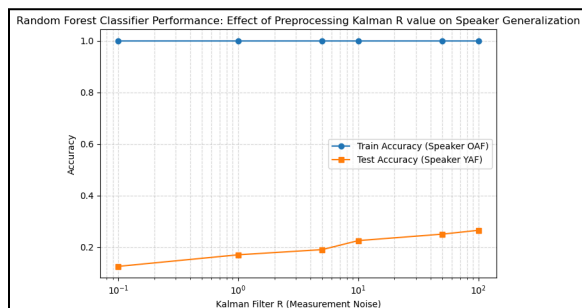
Hyper parameter tuning was completed across C values 0.01, 0.1, 1, 10, and 100 to test for regularization with GridSearchCV identifying the best parameter was a C value of 1 with best CV accuracy of 99.6% on the training dataset. The smaller C value suggests that optimal performance was achieved with softer margins, which aligns with the aim to reduce individual speaker variability in favor of generalization. However, there was a large gap between test and train accuracy, which reached 100% accuracy quickly, suggesting that overfitting occurred.



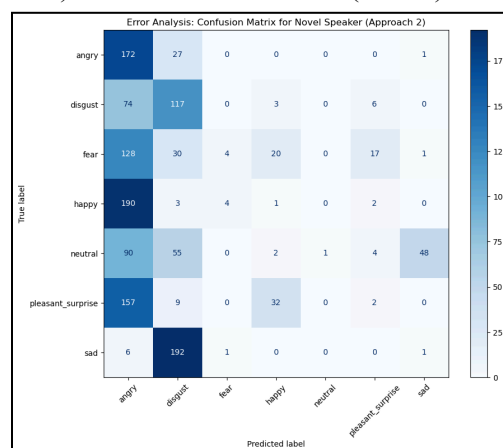
	precision	recall	f1-score	support
angry	0.13	0.06	0.08	200
disgust	0.76	0.24	0.37	200
fear	0.08	0.32	0.13	200
happy	0.00	0.00	0.00	200
neutral	0.00	0.00	0.00	200
pleasant_surprise	0.43	0.85	0.57	200
sad	0.44	0.06	0.10	200
accuracy			0.22	1400
macro avg	0.26	0.22	0.18	1400
weighted avg	0.26	0.22	0.18	1400

The test accuracy when generalizing on a novel speaker was 21.9% and the error rate was 78.1% (1094 out of 1400 younger female speaker test audio files were misclassified). This was slightly better than random chance (14%). Overall, the model has high recall for pleasant surprise, but low precision, suggesting it is often over-guessing this emotion. The highest precision was for disgust, but has low recall, suggesting that while it identified disgust correctly, it was missing a lot of cases. The model also frequently guessed fear but was often wrong, and missed most cases of happy and neutral.

Approach 2: Kalman Smoothing of MFCCs & Deltas + Random Forest



The best Kalman R value is 100 with a test accuracy of 26.5% when generalizing to a subset of young female speakers using a Random Forest classifier with 100 decision trees, max depth of 10, and random state of 42. There was again an issue of overfitting as shown by the gap between train and test accuracy with train accuracy reaching 100% immediately, however there was a notable benefit of smoothing based on the positive trend seen in the test accuracy as the R value becomes larger. Interestingly, test accuracy dropped to 18.1% when introduced to the entire test dataset, therefore further regularization was required with the best test accuracy (22.1%) achieved with an R value of 100, 100 decision trees, max depth of 5, minimum samples per leaf of 20, and random state of 42. MFCC means and standard deviations had higher feature importance (75.2%) than the Delta features (24.7%).

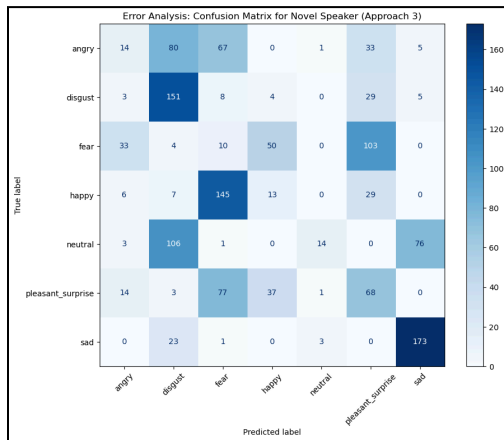


	precision	recall	f1-score	support
angry	0.21	0.86	0.34	200
disgust	0.27	0.58	0.37	200
fear	0.44	0.02	0.04	200
happy	0.02	0.01	0.01	200
neutral	1.00	0.01	0.01	200
pleasant_surprise	0.06	0.01	0.02	200
sad	0.02	0.01	0.01	200
accuracy			0.21	1400
macro avg	0.29	0.21	0.11	1400
weighted avg	0.29	0.21	0.11	1400

The test accuracy when generalizing on a novel speaker was 21.3% and the error rate was 78.7% (1102 out of 1400 younger female speaker test audio files were misclassified). While accuracy was similar to the previous

model, this model had a strong bias towards angry and disgust with otherwise poor performance for all other emotions. Also, while it had a perfect identification of neutral, the recall was 0.01 with only one neutral case even identified.

Approach 3: Kalman Smoothing of MFCCs & Deltas + Multi-Layer Perceptron

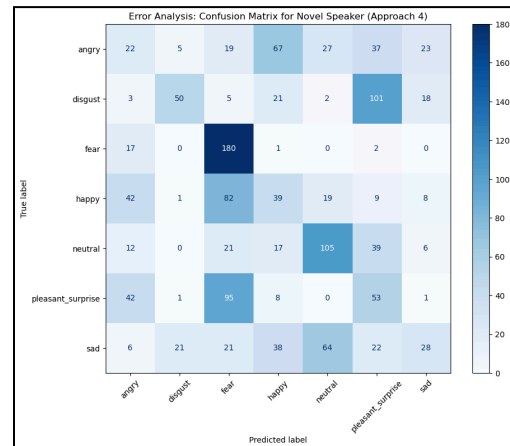


Classification Report (Approach 3)				
	precision	recall	f1-score	support
angry	0.19	0.07	0.10	200
disgust	0.40	0.76	0.53	200
fear	0.03	0.05	0.04	200
happy	0.12	0.07	0.09	200
neutral	0.74	0.07	0.13	200
pleasant_surprise	0.26	0.34	0.29	200
sad	0.67	0.86	0.75	200
accuracy			0.32	1400
macro avg	0.35	0.32	0.28	1400
weighted avg	0.35	0.32	0.28	1400

The test accuracy when generalizing on a novel speaker was 31.6% and the error rate was 68.4% (957 out of 1400 younger female speaker test audio files were misclassified). This was an improvement compared to the previous two models, with macro average F1-score now at 0.28, compared to previous scores of 0.18 for the first model and 0.11 for the second model. This third model has high precision and recall for sad and disgust, suggesting the model is accurately identifying these cases. However, there is still risk that the model is just favoring guessing disgust based on its comparatively low precision. There is also some confusion between angry, disgust, and fear in particular. Similar to the second model, for neutral, this model has the

issue of excellent precision but low recall, so while it may be good at identifying neutral cases, it still misses the majority of them. Lastly, there was overall good precision and recall for pleasant surprise, though pleasant surprise was often misclassified as fear.

Approach 4: wav2vec 2.0 + Multi-Layer Perceptron

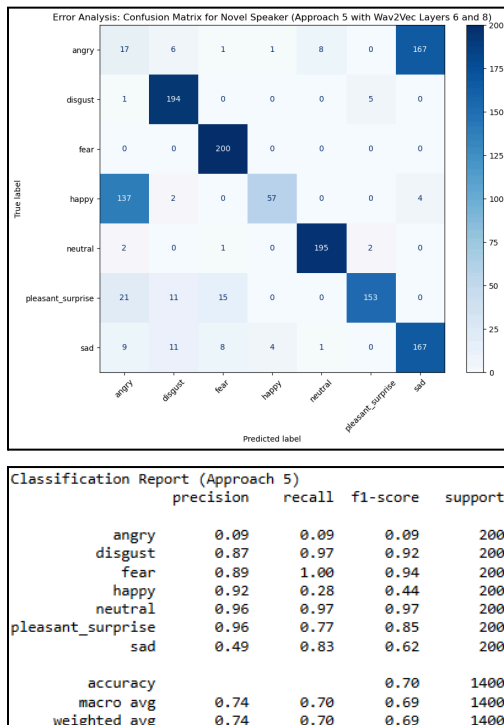


Classification Report (Approach 4)				
	precision	recall	f1-score	support
angry	0.15	0.11	0.13	200
disgust	0.64	0.25	0.36	200
fear	0.43	0.90	0.58	200
happy	0.20	0.20	0.20	200
neutral	0.48	0.53	0.50	200
pleasant_surprise	0.20	0.27	0.23	200
sad	0.33	0.14	0.20	200
accuracy			0.34	1400
macro avg	0.35	0.34	0.31	1400
weighted avg	0.35	0.34	0.31	1400

The test accuracy when generalizing on a novel speaker was 34.1% and the error rate was 65.9% (923 out of 1400 younger female speaker test audio files were misclassified). While this accuracy is only slightly better than the third model, the macro average F1-score improved to 0.31 and six emotions (disgust, fear, happy, neutral, pleasant surprise, sad) achieved F1-scores of 0.20 or greater, suggesting increased consistency in classifying the majority of emotions. While this fourth model more accurately identifies fear, the high recall suggests it still suffers from over-guessing, i.e. false positives. This model continues to perform well with disgust, however it has low recall, thus missing 75% of disgust cases. It also improves

in correctly identifying the neutral cases, and has higher recall than the third model where now less than half of the neutral cases are missed, i.e. false negatives. Lastly, this model does perform worse at classifying angry cases compared to the third model, often misclassifying it as happy or pleasant surprise. Overall, angry, happy, and pleasant surprise have similarly low but balanced precision and recall with F1-scores ranging between 0.13 and 0.23, suggesting that this model is correctly identifying some of these cases but is not learning useful patterns for identifying these emotions.

Approach 5: wav2vec 2.0 with Layer Extraction & Pyramid Pooling + Multi-Layer Perceptron



Results analysis focused on the model that used ensemble layer extraction due to achieving the best performance. The test accuracy when generalizing on a novel speaker using features extracted from hidden state layers 6 and 8 was 70.2%, train accuracy was 99.9% (compared to 50.9% test accuracy and 99.8% train accuracy with layer 8 only), and the error rate was 29.8% (417 out of 1400 younger female

speaker test audio files were misclassified). This was a significant improvement compared to all previous models with a macro average F1-score of 0.69. This model displayed strong generalization to five out of seven emotions, achieving F1-scores of greater than 0.60 for disgust, fear, neutral, pleasant surprise, and sad cases. While precision was high for happy, performance was poor due to low recall, frequently misclassifying happy as angry. This model also struggled with angry cases, with low recall, precision, and F1-score, it essentially did not learn any generalizable features, instead showing a tendency to classify angry as sad.

Discussion

The best performance was achieved using an MLP classifier with extraction of hidden state layers 6 and 8 from wav2vec 2.0 output and further processing using pyramid pooling to generate mean and standard deviation features across the time domain. This success likely comes from wav2vec 2.0's strengths from a transformer-based approach trained across many hours of speaker data and was enhanced by eliminating noise through hand-selecting specific layers. Pyramid pooling also provided data augmentation, thus increasing the quantity of features for training and giving different acoustic angles for the model to train on. All of the models struggled with overfitting, with even the best model still showing about a 30% gap between train and test accuracy.

However the use of functionals derived from static spectral features, like MFCCs and Deltas, had significantly worse performance, even with smoothing via Kalman Filter compared to the convolutional neural network based transformer. Likely because these other approaches learned too closely to individual speaker characteristics, preventing the models from being able to later focus on the speaker invariant emotional features extracted from the raw test audio. As well, MLP models had better

performance compared to less computationally expensive models like C-Support Vector and Random Forest classifiers.

Even the best model had difficulty with classification of happy and angry. False negatives for happy were often misclassified as angry, but angry cases were misclassified as sad, suggesting that paralinguistic acoustic features of these three emotions have overlapping similarities with anger being the hardest for the model to characterize. It is possible that some of these challenges are due to speaker specific influences on emotion interpretation. Limitations of this work and opportunity for future exploration include testing this classifier on a wider variety of speakers and more natural, spontaneous, longer duration speech contexts, as well as more targeted fine-tuning of hidden state layer extraction to optimize transfer learning using the wav2vec 2.0 model.

Analysis Code and Other Documents

https://github.com/rellin01/CS135_FinalProject

Citations

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